

DevOps Practical Assignment (Beginner to Intermediate)

Objective

This assignment provides hands-on experience with core cloud and containerization tools:

- **AWS S3:** Versioning and Static Hosting.
- **Docker:** Custom Image and Docker Hub.
- **AWS EC2:** Nginx Web Server Deployment.

TASK 1: S3 Bucket Versioning + Static Website Hosting

Goal: Create an S3 bucket, enable versioning, and host a static website.

Step 1: Create S3 Bucket

1. Navigate to S3 in the AWS Console.
2. Click "Create Bucket" and enter a unique name.
3. **Disable** "Block all public access".

Step 2: Enable Versioning

1. In bucket **Properties**, enable **Bucket Versioning**.

Step 3: Upload Website Files

1. Create an `index.html` file with the following code:

HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```



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```
<title>My First S3 Website</title>
```

```
</head>
```

```
<body>
```

```
<h1>Hello from S3</h1>
```

```
</body>
```

```
</html>
```

2. Upload this file to your bucket.

Step 4: Enable Static Hosting

1. In **Properties**, enable **Static Website Hosting** and set the index document to `index.html`.

Step 5: Make Files Public

1. Under **Permissions**, add the following **Bucket Policy**:

JSON

```
{  
  
  "Version": "2012-10-17",  
  
  "Statement": [  
  
    {  
  
      "Effect": "Allow",  
  
      "Principal": "*",
```



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```
"Action": "s3:GetObject",  
"Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"  
}  
]  
}
```

Verification: Access the "Bucket Website Endpoint" in your browser. Confirm the site loads and verify versioning by uploading a new `index.html`.

TASK 2: Create Custom Dockerfile and Push to Docker Hub

Goal: Build a Docker image and push it to Docker Hub.

Step 1: Create Project Files

Create a folder with the following two files: **app.py**

Python

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route('/')  
def home():  
    return "Hello from Docker"  
  
if __name__ == '__main__':
```



```
app.run(host='0.0.0.0', port=5000)
```

requirements.txt

Plaintext

flask

Step 2: Create Dockerfile

Dockerfile

```
FROM python:3.9-slim
```

```
WORKDIR /app
```

```
COPY requirements.txt .
```

```
RUN pip install -r requirements.txt
```

```
COPY . .
```

```
CMD ["python", "app.py"]
```

Step 3: Build and Push

1. **Build:** `docker build -t yourusername/myapp:latest .`
2. **Login:** `docker login`
3. **Push:** `docker push yourusername/myapp:latest`

TASK 3: Launch EC2 and Serve Nginx Page

Goal: Launch an EC2 instance and host a webpage using Nginx.



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Step 1: Launch and Connect

1. Launch a **t2.micro** instance (Amazon Linux/Ubuntu).
2. Allow **HTTP (80)** and **SSH (22)** in the Security Group.
3. Connect via SSH: `ssh -i your-key.pem ec2-user@your-public-ip`.

Step 2: Install and Start Nginx

- **For Ubuntu:** `sudo apt update && sudo apt install nginx -y`
- **Start Service:** `sudo systemctl start nginx`

Step 3: Modify Web Page

1. Run: `sudo nano /usr/share/nginx/html/index.html`
2. Replace content with: `<h1>Hello from EC2 Nginx</h1>`

Evaluation Criteria

Criteria	Marks
S3 Setup	30
Docker Task	30
EC2 + Nginx	30



Documentation	10
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Final Deliverables: S3 URL + Screenshot, Docker Hub Link, and EC2 Public IP Screenshot.



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